

Lab 3: Relational Algebra

Query 1

Description: Find the first and last names of persons who have made at least one equipment reservation.

Answer:

- $\pi[\text{Person.firstName}, \text{Person.lastName}] (\text{Person} \bowtie [\text{Person.pid} = \text{Reservation.madeByPid}] \text{Reservation})$

Query 2

Description: Find the IDs of persons who are either attached to a laboratory or act as PI of at least one research project.

Answer:

- $\pi[\text{Attached.pid}] (\text{Attached}) \cup \pi[\text{Participates.pid}] (\sigma[\text{Participates.role} = \text{'PI'}] (\text{Participates}))$

Query 3

Description: Find the IDs of laboratories located in the building 'College of Computing' or containing at least one equipment unit whose model belongs to the category 'CPU Server'.

Answer:

- $\pi[\text{Laboratory.labId}] (\sigma[\text{Laboratory.building} = \text{'College of Computing'}] (\text{Laboratory})) \cup \pi[\text{EquipmentUnit.labId}] (\text{EquipmentUnit} \bowtie [\text{EquipmentUnit.modelId} = \text{EquipmentModel.modelId}] \sigma[\text{EquipmentModel.category} = \text{'CPU Server'}] (\text{EquipmentModel}))$

Query 4

Description: Find the IDs of laboratories that contain at least one equipment unit whose model belongs to the category 'CPU Server' and at least one equipment unit whose model belongs to the category 'GPU Server'.

Answer:

- $A = \pi[\text{EquipmentUnit.labId}] (\text{EquipmentUnit} \bowtie [\text{EquipmentUnit.modelId} = \text{EquipmentModel.modelId}] \sigma[\text{EquipmentModel.category} = \text{'CPU Server'}] (\text{EquipmentModel}))$
- $B = \pi[\text{EquipmentUnit.labId}] (\text{EquipmentUnit} \bowtie [\text{EquipmentUnit.modelId} = \text{EquipmentModel.modelId}] \sigma[\text{EquipmentModel.category} = \text{'GPU Server'}] (\text{EquipmentModel}))$
- $\text{Result} = A \cap B$

Query 5

Description: Find the Faculty members who supervise every laboratory located in the building 'College of Computing'.

Answer:

- $CC = \pi[\text{Laboratory.labId}] (\sigma[\text{Laboratory.building} = \text{'College of Computing'}] (\text{Laboratory}))$
- $\text{Supervision} = \rho[\text{Supervision(pid, labId)}]$
 $(\pi[\text{Laboratory.supervisorPid, Laboratory.labId}] (\text{Laboratory}))$
- $\text{Qualified} = \text{Supervision} \div CC$
- $\text{Result} = \pi[\text{Faculty.pid, Faculty.position}] (\text{Faculty} \bowtie [\text{Faculty.pid} = \text{Qualified.pid}] \text{Qualified})$

Query 6

Description: Find the IDs of persons who participate in every research project funded by at least one budget that also funds the laboratory with labId = 2.

Answer:

- $A = \pi[\text{Participates.pid}, \text{Participates.projectCode}] (\text{Participates})$
- $\text{Funds} = \pi[\text{FundsLab.budgetLine}] (\sigma[\text{FundsLab.labId} = 2] (\text{FundsLab}))$
- $B = \pi[\text{FundsPrj.projectCode}] (\text{FundsPrj} \bowtie [\text{FundsPrj.budgetLine} = \text{Funds.budgetLine}] \text{Funds})$
- $\text{Result} = A \div B$

Query 7

Description: Find pairs of persons (p1, p2) such that p1 has made more reservations than p2.

Answer:

- $C = \rho[C(\text{pid}, \text{cnt})] (\text{madeByPid } G [\text{COUNT}(\text{resId}) \rightarrow \text{cnt}] (\text{Reservation}))$
- $\text{No_res} = \pi[\text{Person.pid}] (\text{Person}) - \pi[C.\text{pid}] (C)$
- $\text{Total} = C \cup (\text{No_res} \times \{(\emptyset)\})$
- $A = \rho[A(\text{pid1}, \text{cnt1})] (\text{Total})$
- $B = \rho[B(\text{pid2}, \text{cnt2})] (\text{Total})$
- $\text{Result} = \pi[A.\text{pid1}, B.\text{pid2}] (A \bowtie [A.\text{cnt1} > B.\text{cnt2}] B)$

Query 8

Description: Find the IDs of persons who have reserved equipment units located in at least two different laboratories.

Answer:

- $R = \pi[\text{Reservation.madeByPid}, \text{EquipmentUnit.labId}] (\text{Reservation} \bowtie [\text{Reservation.resId} = \text{Reserves.resId}] \text{Reserves} \bowtie [\text{Reserves.serialNo} = \text{EquipmentUnit.serialNo}] \text{EquipmentUnit})$
- $R1 = \rho[R1(\text{pid}, \text{lab1})] (R)$
- $R2 = \rho[R2(\text{pid}, \text{lab2})] (R)$

- $\text{Result} = \pi[R1.pid] (R1 \bowtie [R1.pid = R2.pid \wedge R1.lab1 \neq R2.lab2] R2)$

Query 9

Description: Find the IDs of reservations submitted in September 2026 that reserve at least one equipment unit currently located in the laboratory named 'AI Movement'.

Answer:

- $A = \pi[\text{Reservation.resId}] (\sigma[\text{Reservation.submissionTS} \geq '2026-09-01' \wedge \text{Reservation.submissionTS} < '2026-10-01'] (\text{Reservation}))$
- $B = \sigma[\text{Laboratory.name} = 'AI Movement'] (\text{Laboratory})$
- $\text{Result} = \pi[A.resId] (A \bowtie [A.resId = \text{Reserves.resId}] \text{Reserves} \bowtie [\text{Reserves.serialNo} = \text{EquipmentUnit.serialNo}] \text{EquipmentUnit} \bowtie [\text{EquipmentUnit.labId} = B.labId] B)$

Query 10

Description: Find the IDs of persons who act as PI of more than one research project.

Answer:

- $R = \pi[\text{Participates.pid}, \text{Participates.projectCode}] (\sigma[\text{Participates.role} = 'PI'] (\text{Participates}))$
- $R1 = \rho[R1(pid, code1)] (R)$
- $R2 = \rho[R2(pid, code2)] (R)$
- $\text{Result} = \pi[R1.pid] (R1 \bowtie [R1.pid = R2.pid \wedge R1.code1 \neq R2.code2] R2)$

Query 11

Description: Find the IDs of approvers who have approved reservations involving equipment units located in more than one laboratory.

Answer:

- $A = \pi[\text{Reservation.approverPid}, \text{Reservation.resId}]$
 $(\sigma[\text{Reservation.status} = \text{'Approved'} \wedge \text{Reservation.approverPid is not NULL}] (\text{Reservation}))$
- $R = \pi[A.approverPid, \text{EquipmentUnit.labId}] (A \bowtie [A.resId = \text{Reserves.resId}] \text{Reserves} \bowtie [\text{Reserves.serialNo} = \text{EquipmentUnit.serialNo}] \text{EquipmentUnit})$
- $R1 = \rho[R1(pid, lab1)] (R)$
- $R2 = \rho[R2(pid, lab2)] (R)$
- $\text{Result} = \pi[R1.pid] (R1 \bowtie [R1.pid = R2.pid \wedge R1.lab1 \neq R2.lab2] R2)$

Query 12

Description: Find the IDs of persons who have no reservation whose planned reservation interval includes November 6, 2026.

Answer:

- $C = \pi[\text{Reservation.madeByPid}] (\sigma[\text{Reservation.plannedStart} \leq \text{'2026-11-06 23:59:59'} \wedge \text{Reservation.plannedEnd} \geq \text{'2026-11-06 00:00:00'}] (\text{Reservation}))$
- $\text{Result} = \pi[\text{Person.pid}] (\text{Person}) - \rho[C1(pid)] (C)$

Query 13

Description: Find the laboratories whose number of equipment units is below the average number of equipment units per laboratory.

Answer:

- $A = \rho[A(\text{labId}, \text{cnt})] (\text{EquipmentUnit.labId } G [\text{COUNT}(\text{EquipmentUnit.serialNo}) \rightarrow \text{cnt}] (\text{EquipmentUnit}))$
- $\text{No_unit} = \rho[\text{No_unit}(\text{labId}, \text{cnt})] ((\pi[\text{Laboratory.labId}] (\text{Laboratory}) - \pi[A.labId] (A)) \times \{(\emptyset)\})$

- $Total = A \cup No_unit$
- $B = G [AVG(No_unit.cnt) \rightarrow avg_cnt] (Total)$
- $Result = \pi[Total.labId] (Total \bowtie [Total.cnt < B.avg_cnt] B)$

Query 14

Description: For each laboratory, return the person who has made the greatest number of distinct approved reservations involving equipment units located in that laboratory.

Answer:

- $A = \pi[Reservation.resId, Reservation.madeByPid] (\sigma[Reservation.status = 'Approved'] (Reservation))$
- $B = \pi[EquipmentUnit.labId, A.madeByPid, A.resId] (A \bowtie [A.resId = Reserves.resId] Reserves \bowtie [Reserves.serialNo = EquipmentUnit.serialNo] EquipmentUnit)$
- $C = \rho[C(labId, pid, cnt)] (B.labId, B.madeByPid G [COUNT(B.resId) \rightarrow cnt] (B))$
- $D = \rho[D(labId, max_cnt)] (C.labId G [MAX(C.cnt) \rightarrow max_cnt] (C))$
- $Result = \pi[C.labId, C.pid] (C \bowtie [C.labId = D.labId \wedge C.cnt = D.max_cnt] D)$

Query 15

Description: List the IDs of persons who made at least three equipment reservations during the year 2025.

Answer:

- $A = \sigma[Reservation.submissionTS \geq '2025-01-01' \wedge Reservation.submissionTS < '2026-01-01'] (Reservation)$
- $B = \rho[B(pid, cnt)] (A.madeByPid G [COUNT(A.resId) \rightarrow cnt] (A))$
- $Result = \pi[B.pid] (\sigma[B.cnt \geq 3] (B))$